

# ODCalc

## Terms and Conditions of Use

Last Updated: March 2026

---

**IMPORTANT:** Please read these Terms and Conditions carefully before using ODCalc. By downloading, installing, or using this application, you acknowledge that you have read, understood, and agree to be bound by these terms. If you do not agree with these terms, do not use this application.

### 1. Acceptance of Terms

These Terms and Conditions ("Terms") constitute a legally binding agreement between you (the "User") and Dr. PA Gallardo Molina ("Developer", "we", "us", or "our") regarding your use of the ODCalc mobile application ("App", "Application", or "Software").

Your use of the App constitutes your acceptance of these Terms. We reserve the right to modify these Terms at any time, and such modifications will be effective immediately upon posting. Your continued use of the App following any changes indicates your acceptance of the new Terms.

### 2. License Grant and Restrictions

Subject to your compliance with these Terms, we grant you a limited, non-exclusive, non-transferable, revocable license to download, install, and use the App on devices that you own or control, solely for your personal, educational, or professional microbiological research purposes.

You may NOT:

- Modify, reverse engineer, decompile, or disassemble the App
- Remove, alter, or obscure any proprietary notices from the App
- Use the App for any illegal, unauthorized, or prohibited purpose
- Distribute, sublicense, rent, lease, or lend the App to third parties
- Create derivative works based on the App
- Use the App in any manner that could damage, disable, overburden, or impair the App

### 3. Intended Use and Scope

ODCalc is designed as an educational and research tool for calculating bacterial culture dilutions based on user-provided optical density (OD) measurements. The App is intended to:

- Assist in routine microbiological laboratory calculations
- Estimate the volume of diluent required to achieve a target OD from a measured OD
- Support educational purposes in microbiology and related disciplines
- Facilitate research and development activities in laboratory settings
- Provide preliminary, approximated dilution guidance

**CRITICAL LIMITATION:** ODCalc relies on the assumption of a linear relationship between OD and cell density (Beer-Lambert law). This assumption does not hold strictly at high cell densities, in cultures with turbidity contributions from non-cellular components, or across different spectrophotometric configurations. Results are approximations and must be verified experimentally. This App is NOT intended for clinical diagnostic purposes.

medical decision-making, or any application where accuracy is critical for patient safety or public health decisions.

#### 4. Disclaimer of Warranties

THE APP IS PROVIDED "AS IS" AND "AS AVAILABLE" WITHOUT WARRANTIES OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO: WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, OR NON-INFRINGEMENT; WARRANTIES THAT THE APP WILL BE UNINTERRUPTED OR ERROR-FREE; WARRANTIES REGARDING THE ACCURACY, RELIABILITY, OR COMPLETENESS OF CALCULATION RESULTS; OR WARRANTIES THAT DEFECTS WILL BE CORRECTED OR THAT UPDATES WILL BE PROVIDED.

We do not warrant that:

- Dilution calculation results will be accurate, complete, or reliable for all experimental conditions
- The App will be compatible with all devices or operating system versions
- Results will be consistent across different spectrophotometers, pathlength configurations, or culture conditions
- Any data stored in the App will be preserved or accessible

YOU ACKNOWLEDGE AND AGREE THAT YOUR USE OF THE APP IS AT YOUR SOLE RISK.

#### 5. Limitation of Liability

TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, IN NO EVENT SHALL THE DEVELOPER BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL, OR EXEMPLARY DAMAGES; LOSS OF PROFITS, REVENUE, DATA, OR BUSINESS OPPORTUNITIES; INACCURATE DILUTION CALCULATIONS OR ANALYTICAL RESULTS; DECISIONS OR ACTIONS TAKEN BASED ON APP RESULTS; HARM RESULTING FROM INCORRECT, INCOMPLETE, OR UNRELIABLE RESULTS; OR INCOMPATIBILITY WITH DEVICES, OPERATING SYSTEMS, OR OTHER SOFTWARE. THIS LIMITATION APPLIES REGARDLESS OF THE LEGAL THEORY UPON WHICH THE CLAIM IS BASED, EVEN IF WE HAVE BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

If you are dissatisfied with the App or these Terms, your sole and exclusive remedy is to discontinue use of the App.

**MAXIMUM LIABILITY:** In any case, our total aggregate liability shall not exceed the amount you paid for the App (if any), or \$10 USD, whichever is greater.

#### 6. Scientific Accuracy and Inherent Limitations

ODCalc uses the Beer-Lambert law as a mathematical basis for dilution calculations. Users must be aware of the following inherent limitations:

- OD-to-cell-density linearity is typically valid only within a limited OD range (commonly  $OD_{600} < 0.3-0.5$ , depending on instrument and organism)
- At higher optical densities, light scattering and multiple scattering effects cause non-linear deviations that result in underestimation of true cell density
- Results depend on the spectrophotometer model, cuvette pathlength, wavelength calibration, and instrument-specific correction factors
- Culture medium turbidity, pigmentation, or particulate matter may confound OD readings independently of cell density
- Calculated dilution factors are approximations and must be validated experimentally for each organism and culture condition

YOU ACKNOWLEDGE THAT AUTOMATED CALCULATIONS ARE INHERENTLY SUBJECT TO ERROR AND THAT YOU WILL VERIFY ALL CRITICAL RESULTS USING STANDARD LABORATORY METHODS AND PROFESSIONAL JUDGMENT.

## 7. User Responsibilities

You are solely responsible for:

- **Verifying all results:** You must independently verify all dilution calculations and results generated by the App using validated laboratory procedures and professional scientific judgment
- **Data backup:** Creating and maintaining backups of any important data. We are not responsible for any data loss
- **Proper use:** Using the App in accordance with good laboratory practices (GLP) and applicable institutional or regulatory guidelines
- **Regulatory compliance:** Ensuring your use of the App complies with all applicable laws, regulations, and institutional policies
- **Device security:** Maintaining the security of your device and protecting access to the App
- **Professional judgment:** Exercising appropriate professional and scientific judgment when interpreting and applying results

## 8. No Medical or Diagnostic Use

**CRITICAL NOTICE:** This App is NOT a medical device and is NOT intended for use in the diagnosis, treatment, cure, mitigation, or prevention of disease. It has NOT been evaluated or approved by the FDA, CE, or any other regulatory authority. It should NOT be used for clinical decision-making and is NOT intended for use in patient care or public health surveillance. All results must be verified using validated, approved methods before being used for any critical purpose.

## 9. In-App Purchases (Apple Watch Version)

The iOS version of ODCalc is provided free of charge. A paid version is available for Apple Watch. If you make a purchase to unlock the Apple Watch version:

- Purchase transactions are handled entirely by Apple through the App Store
- We do not have access to your payment information
- Apple's Privacy Policy governs the collection and use of payment data
- All other Terms remain in full force and effect for both versions

## 10. Data Loss and Backup

WE ARE NOT RESPONSIBLE FOR ANY DATA LOSS, CORRUPTION, OR UNAVAILABILITY.

You acknowledge that:

- All data is stored locally on your device
- Data may be lost due to device failure, app errors, OS updates, or other causes
- Uninstalling the App will delete all locally stored data
- App updates may cause data incompatibility or loss
- We do not provide cloud backup or data recovery services
- You are solely responsible for backing up any important data

## **11. Updates and Support**

We may, but are not obligated to:

- Provide updates, bug fixes, or improvements to the App
- Maintain compatibility with future device models or OS versions
- Provide technical support or assistance
- Continue offering the App indefinitely

We reserve the right to discontinue the App or any features at any time without notice or liability.

## **12. Intellectual Property**

All intellectual property rights in and to the App, including but not limited to software code, algorithms, design, user interface, documentation, and trademarks, are owned by or licensed to the Developer. Your use of the App does not grant you any ownership rights to the App or its content.

## **13. Third-Party Components**

The App may incorporate third-party software components, which are subject to their own licenses and terms. We are not responsible for any third-party components or their performance.

## **14. Indemnification**

You agree to indemnify, defend, and hold harmless the Developer from and against any and all claims, damages, losses, costs, expenses, and liabilities (including reasonable attorneys' fees) arising from:

- Your use or misuse of the App
- Your violation of these Terms
- Your violation of any rights of another party
- Any decisions or actions taken based on App results

## **15. Governing Law and Jurisdiction**

The App is available worldwide. These Terms shall be governed by and construed in accordance with the laws of Chile, without regard to its conflict of law provisions. Any disputes arising from these Terms or your use of the App shall be subject to the exclusive jurisdiction of the courts of Chile.

## **16. Severability**

If any provision of these Terms is found to be invalid, illegal, or unenforceable, the remaining provisions shall continue in full force and effect.

## **17. Entire Agreement**

These Terms, together with our Privacy Policy, constitute the entire agreement between you and the Developer regarding your use of the App and supersede all prior agreements and understandings.

## **18. No Waiver**

Our failure to enforce any right or provision of these Terms will not be considered a waiver of those rights.

## **19. Termination**

We may terminate or suspend your access to the App immediately, without prior notice or liability, for any reason, including if you breach these Terms. Upon termination, your right to use the App will immediately cease, and you must delete all copies of the App from your devices.

## **20. Export Compliance**

You agree to comply with all applicable export and import laws and regulations in your use of the App.

## **21. Contact Information**

For questions regarding these Terms and Conditions, please contact:

**Email:** [pagallardo@ug.uchile.cl](mailto:pagallardo@ug.uchile.cl)

**Developer:** Dr. PA Gallardo Molina

---

BY USING ODCALC, YOU ACKNOWLEDGE THAT YOU HAVE READ THESE TERMS AND CONDITIONS, UNDERSTAND THEM, AND AGREE TO BE BOUND BY THEM.